FORWARD DEPLOYED ENGINEERING

Lec-01

Course Orientation: AI Workers, Certification, Plugins, and Modern Cloud Workflows

1) The plan overview (3 months): certification first, then plugins/products

The plan is shared as a sequence:

- **Step 1: Certification** to establish internationally recognized credibility.
- **Step 2: Build products** (specifically plugins).
- **Step 3: Make products sellable internationally** without spending huge amounts.

2) Why Anthropic certifications are targeted

The instructors identify two leading AI entities discussed in the session: **Entropic** and **OpenAI**. They focus on Entropic's recently announced certifications that are described as **proprietary and practical** rather than a traditional exam-style system.

The certification process is described as: prepare according to the syllabus, then take a certification attempt. There is a proctor on the other side. If students pass, they receive a certification and badge. The badge is framed as a "key" that helps access international opportunities.

3) "Agent Factory" book and navigation to certification sections

The course uses a book as the reference source: **Agent Factory**. The left sidebar contains sections including crash courses and, crucially, certification proof/material.

The instructors explain that Entropic announced four certification levels. These include beginner-to-higher tiers, intended for both technical and non-technical professionals as long as they can use AI properly (not just "tips and tricks").

4) Which certification the course focuses on

The main course focus is on **two certifications** described in the session:

- **Gold Certified Associate Foundations** (for everyone): The exam tests whether you truly know how to use Claude/AI tools to generate real value and solve business problems—not just theory.
- **Cloud/Architect-focused certification** (presented as the "core focus"): They describe it as enabling students to architect AI solutions and apply AI sessions for businesses.

The Associate-focused exam fee is stated as **$125**. The instructors warn that failing wastes part of the fee (they mention $125 being wasted if you fail).

5) Internal exam before the official certification

Before taking the official exam, students take an **internal exam**. The internal exam is described as tougher than Anthropic's, designed to predict pass probability. Passing the internal exam qualifies

students to proceed to the official certification attempt since the team is partnering with Anthropic/Entropic.

Easy Notes: Plugins, Marketplaces, KSOR & DSOR

6. Plugins — Second Major Pillar

The course has three main steps:

Certification → Build Products/Plugins → Sell Internationally

Why Plugins?

- **Certification** gives international credibility.
- **Plugins** become the actual products you can sell.
- Plugins can be listed on marketplaces.
- Marketplaces can provide access to an existing international audience.

7. What is a Plugin?

A plugin is a **packaged AI capability** that can be installed and reused.

The course explains that plugins can combine:

- **Connectors** → Connect AI to external tools/services.
- **Skills** → Define repeatable operations and workflows.
- **Deterministic behavior** → Make the AI perform tasks in a consistent way.

Simple Example

Connector + Skill + Workflow = Plugin

Once installed, a team can repeatedly use the plugin for a specific task instead of building the workflow again.

The course plans to teach students how to:

1. Build plugins
2. Test plugins
3. Publish plugins
4. Make them available through marketplaces

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8. “Piggy Bank” Marketplace Strategy

The instructors use **“piggy bank”** as a metaphor for plugins.

Main Idea

If you build a plugin but keep it completely outside major marketplaces:

- Fewer people may discover it.
- You have to find users yourself.
- Marketing becomes harder.

If you publish it through an established marketplace:

- Users can discover your plugin.
- The platform already has an audience.
- You can use the marketplace's existing ecosystem.

Simple Formula

Build Plugin → Publish on Marketplace → Get Discovery → Get Users → Earn

The session also mentions that OpenAI has a similar ecosystem concept for making products available to users.

9. KSOR and DSOR

The course introduces two technologies for managing AI workers:

KSOR

Knowledge System of Record

DSOR

Data System of Record

These are designed to help AI workers work with reliable knowledge and data.

10. What is a System of Record (SOR)?

A **System of Record** is a trusted source that contains the information an AI worker should rely on.

Simple Example: Exam Preparation

Imagine students have an exam but **no official syllabus**.

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Everyone studies different material.

Result:

- Different preparation
- Different information
- Unpredictable results

Now imagine there is an **official, validated syllabus** containing everything required for the exam.

Students can focus on that trusted material.

Same Idea for AI

AI can sometimes generate answers using:

- Pre-trained knowledge
- Internet information
- Different sources

Some information may be incorrect or unsuitable.

A **System of Record** provides the AI worker with a **trusted reference source**.

11. KSOR — Knowledge System of Record

KSOR = Knowledge System of Record

KSOR is presented as the **authoritative knowledge and policy layer** for AI workers.

It helps define:

- What the AI should know
- Which information it should use
- Which policies it should follow
- How it should perform tasks

Traditional Organization

Traditionally, organizations use:

Human Experts
→ Strategy, decisions, procedures, actions

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Software
→ Execution, tracking, metrics, records

AI Worker Organization

With AI workers:

Human Knowledge + Policies → KSOR → AI Worker

The AI worker becomes a **first-class user** of the organization's knowledge system.

Therefore, organizational knowledge and policies need to be written in a format that an AI worker can reliably understand and follow.

Easy Notes: DSOR, MCP, Agent Skills & AI Workers

11. DSOR — Data System of Record

DSOR = Data System of Record

DSOR is the **governance layer** that controls how an AI worker interacts with real systems.

It defines:

- 🔗 **Connections** — Which systems the AI can connect to
- ✅ **Permissions** — What actions the AI is allowed to perform
- 📝 **Recording** — How action results are stored/recorded
- 🚫 **Refusals** — What happens when an action is not permitted

Simple Example

An AI worker may be allowed to:

Read customer data ✅
Create an invoice ✅
Delete customer records ❌

DSOR helps enforce these rules.

Easy Formula

KSOR = What the AI should know/follow

DSOR = What the AI can do with real systems

12. Industry Standards

The session introduces two important concepts used in modern AI systems:

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1. **MCP — Model Context Protocol**
2. **Agent Skills**

13. MCP — Model Context Protocol

MCP = Model Context Protocol

MCP provides a **standardized way for AI systems to connect with external resources and tools**.

Simple Example

Suppose a company has internal documentation.

Instead of retraining the AI model on all the documentation, MCP can allow the AI to:

AI → MCP → Company Documentation/System → Information → AI → Answer

The AI can call external functions/resources and use the returned information to generate a natural-language response.

Easy Definition

MCP is a standard way for AI to access external context, resources, and tools.

Remember

MCP = Connection

It helps AI **connect to outside information and tools**.

14. Agent Skills

Skills = Instructions for repeatable tasks

A skill defines **how an AI should perform a particular task**.

Without Skills

The same task might produce:

- Different processes
- Different outputs
- Different wording
- Less predictable behavior

With Skills

You provide:

- Instructions
- Procedures
- Rules
- Constraints

The AI can then perform the task in a more **consistent and repeatable** way.

Team Benefit

Skills also help teams stay consistent.

Instead of every employee developing their own way of doing something:

One Skill → Same procedure → Entire team

Easy Definition

Skills make AI behavior more consistent for repeatable tasks.

Remember

MCP = Connect

Skills = Perform consistently

15. AI Workers & Virtual Machines

A major concept introduced in the session is the transition from **simple AI chat → AI workers**.

Traditional AI Chat

You:

Give AI a prompt

AI:

Gives you an answer

AI Worker

You:

Give AI a task

AI worker:

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Plans → Runs commands → Uses tools → Performs work → Produces results

Virtual Machine (VM)

The session explains that an AI worker can run inside a **virtual machine** for a session.

The VM may have its own:

- CPU
- RAM
- Disk
- Processes
- Command-line environment

The AI worker can execute commands and perform tasks inside this environment.

Important Concept

The VM is **provided by the cloud/service provider**, but it is allocated for your session.

So the AI worker isn't just generating text—it can have an **actual computing environment in which it performs work**.

16. Cloud Session Advantage

According to the session, because the AI worker operates in the cloud:

Your laptop → Starts the task

↓

Cloud VM → AI worker continues working

↓

You can close/shut down your laptop

↓

Cloud session can continue running

↓

You return later → Retrieve the results

This is different from traditional local tools that stop when your computer shuts down.

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16. Chatbot vs AI Worker

Chatbot Behavior

A chatbot mainly **tells you what to do**.

Example:

“Open the terminal, run this command, and then create the file.”

The AI explains the steps, but **you perform them**.

AI Worker Behavior

An AI worker **does the work on your behalf**.

It can:

- Plan the task
- Use tools
- Use connectors
- Use skills
- Execute commands
- Create files
- Return the final output

Easy Difference

Chatbot → Tells you how

AI Worker → Does it for you

17. Browser, Desktop & Mobile

The course explains that the same AI worker session can be accessed across different devices.

Web

- Provides the browser interface.
- Connects to the cloud session/VM.

Desktop

- Provides deeper access to local files and computer workflows.

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Mobile

- Can access/control the same session.

Main Idea

One account → Same session → Multiple devices

Artifacts and session-related outputs can remain accessible when switching devices.

18. AI Worker Sessions & Virtual Machines

Each AI worker session can have its own **VM (Virtual Machine)**.

Important Properties

- Sessions are isolated.
- Multiple sessions can run at the same time.
- One session's temporary files don't automatically appear in another session.
- Different sessions may receive similar VM configurations.
- Sessions can work in parallel without interfering with each other.

Example

Session A

- Creates project.zip
- Has its own VM/files

Session B

- Starts separately
- Does not automatically have Session A's temporary files

19. Artifacts vs Temporary Files

This is an important distinction.

Artifacts

Permanent/durable outputs

Examples:

- Documents
- Generated project files
- Final deliverables
- Exported files

Artifacts can be:

- Downloaded
- Exported
- Stored
- Used in later work

Temporary Files

Files used only during the worker's session.

When the session ends or VM is destroyed:

Temporary work → Can disappear

Simple Formula

Temporary Work = Session-dependent

Artifact = Durable Output

20. Why Sessions End

Keeping cloud VMs running permanently would consume resources and increase costs.

Therefore:

Start Session → VM Runs → Work Happens → Session Times Out → VM Can Be Destroyed

The important outputs should therefore be saved as **artifacts** before the temporary environment disappears.

21. Session State

Session state refers to the temporary environment and work associated with a particular session.

A worker may use its VM for:

- Running commands

- Creating temporary files
- Processing data
- Testing code
- Performing computations

But the final results should be converted into **durable artifacts**.

22. UI Basics

The course also explains several interface concepts.

Model Selection

Users can select different AI models.

The session mentions model names such as:

- Fable
- Opus
- Sonnet
- Ho

Hardness

"Hardness" is described as relating to the **capability/effort level** used for a request.

Routing

Routing determines how the request is handled.

A request may be routed toward:

- **Chat**
- **AI Worker**
- **Coworker**

Controls

The interface also provides different permission controls.

23. Permissions

There are two important permission modes:

Manual Approval

The AI asks you before performing actions that require permission.

AI → Ask User → User Approves → Action

This gives you more control.

Auto Approval

The system can automatically approve permitted actions.

AI → Action

Course Recommendation

The instructors suggest starting with **manual approval**.

Once you understand and trust the workflow, you can gradually increase automation.

24. Plus Menu / Tools

The interface can provide options such as:

- 📎 Attach files
- 🖼️ Attach images
- 🔌 Enable connectors
- 🌐 Web search
- 🔎 Research options
- ⚙️ Model effort/settings

These tools expand what the AI worker can do.

25. FD 1001 vs FD 102

An important course clarification:

Course Focus

FD 1001 Associate Certification

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Course Focus

FD 102 Architect Certification

FD 1001

Focuses primarily on preparing for the **Associate certification exam**.

FD 102

Focuses on the **Architect certification exam**.

The instructors describe FD 102 as an **advanced/fast-track course**, with emphasis on:

- AI Workers
- Agents
- Plugins
- AI solution architecture

⭐ **Quick Revision**

Chatbot → Tells you

AI Worker → Does the work | Uses the trusted knowledge/data to perform work

VM → Computing environment for the worker

Session → Temporary working environment

Artifact → Durable final output

Temporary file → Can disappear with the session

Manual approval → User approves actions

Auto approval → Actions can happen automatically

FD 1001 → Associate

FD 102 → Architect

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Term	Easy Meaning
Plugin	Packaged AI capability that can be reused
Connector	Connects AI to external tools/services
Skill	Repeatable AI operation/workflow
Marketplace	Platform where users can discover/install plugins
SOR	Trusted source of information
KSOR	Trusted knowledge and policy source for AI workers
DSOR	Trusted data source for AI workers
VM	**Cloud computer/environment where the AI worker can execute tasks**
Cloud Session	**Work can continue on the provider's infrastructure**